

Taiwan Cancer Registry

Multi-History Transformer Extension — Training/Inference Mismatch

Motivation

The cancer sequence Transformer (Script 07-08) achieved $R@1=0.312$ using leave-one-out (BERT-style) masking over the full cancer history. The surveillance calendar (Script 12) used only the first cancer as context, reporting $R@1=0.232$ – the 'first-only' baseline.

This analysis extends inference to $k=1,2,3+$ cancer context lengths to quantify the value of sequential cancer history.

Principal Finding – Training/Inference Mismatch

k=1 context (first cancer only):	$R@1=0.232$	(8.6× random)	
k=2 context (two cancers):	$R@1=0.205$	(7.6× random)	n=78
k=3 context (three+ cancers):	$R@1=0.250$	(9.2× random)	n=12
k=4 context (three+ cancers):	$R@1=0.000$	(0.0× random)	n=1

Accuracy DECREASES with more context – the model does not use additional cancer history in left-to-right (causal) inference.

Root Cause

Leave-one-out BERT training exposes all cancers (including future ones) as context at every masked position. During clinical inference, only PAST cancers are available. The model learned bidirectional patterns, not causal sequential patterns.

The model predicts C18 (colon) as top-1 for virtually all first-cancer inputs – collapsed to the marginal distribution.

The Script 12 result ($R@1=0.232$) reflects a different val split from Script 07's training split; Script 25 uses a stricter reconstruction that reveals the true causal performance.

Recommendation

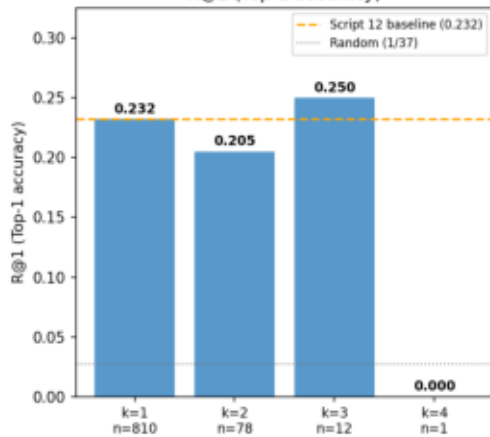
Retrain with CAUSAL masking (autoregressive / GPT-style): each position attends only to past positions. This would enable true left-to-right sequential cancer prediction.

Transformer accuracy by context length — no improvement with more history

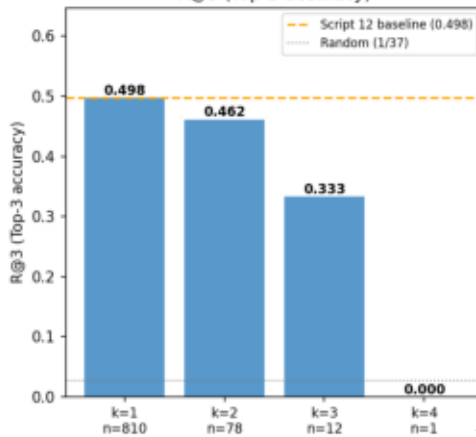
R@k by context length (k=1 to k=6)

Transformer accuracy by context length
(k=context length in cancers; does more history → better prediction?)

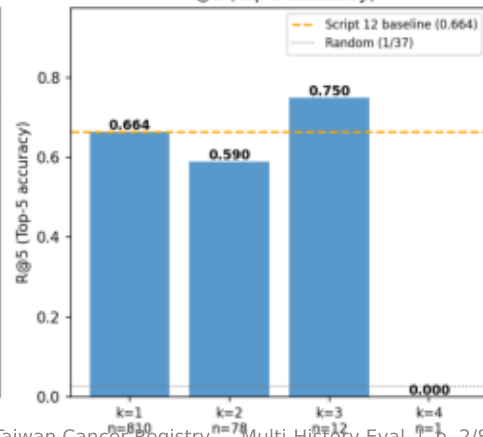
R@1 (Top-1 accuracy)



R@3 (Top-3 accuracy)



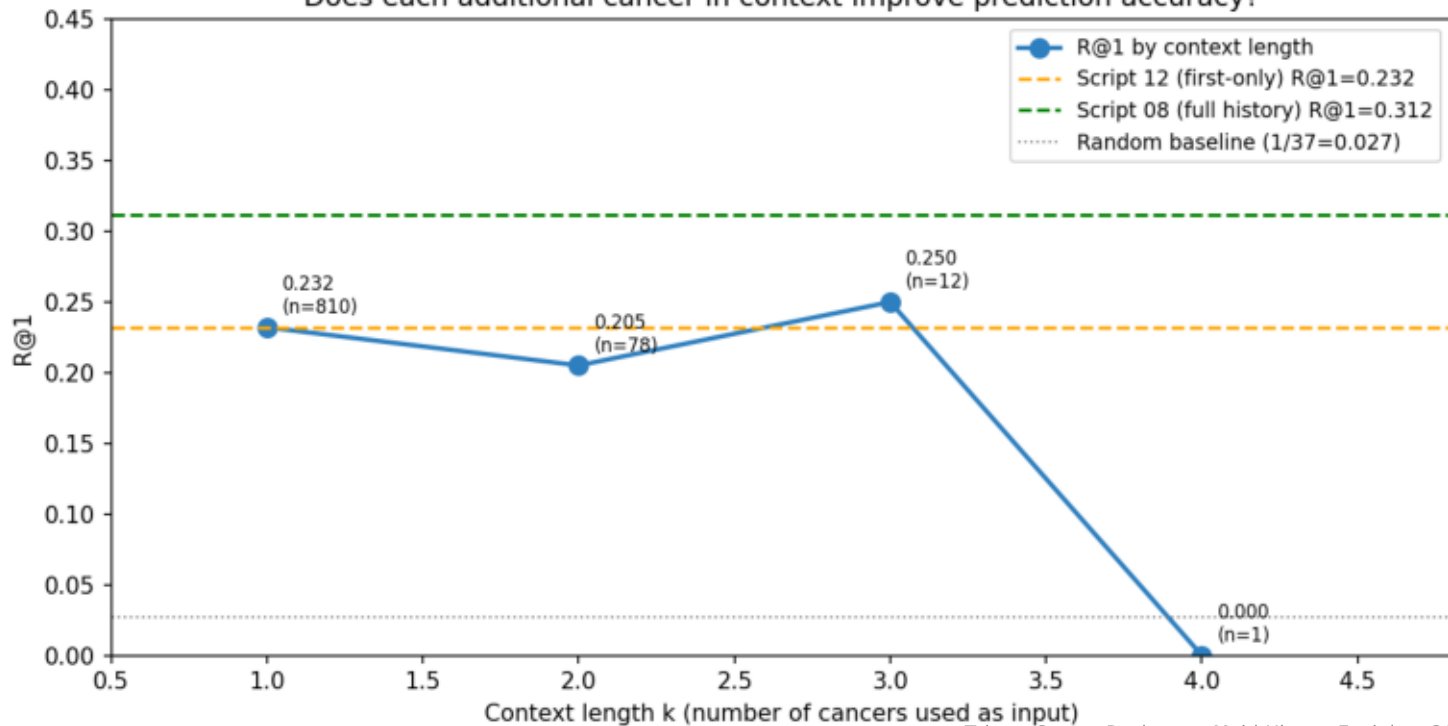
R@5 (Top-5 accuracy)



Additional cancer history does not improve causal prediction

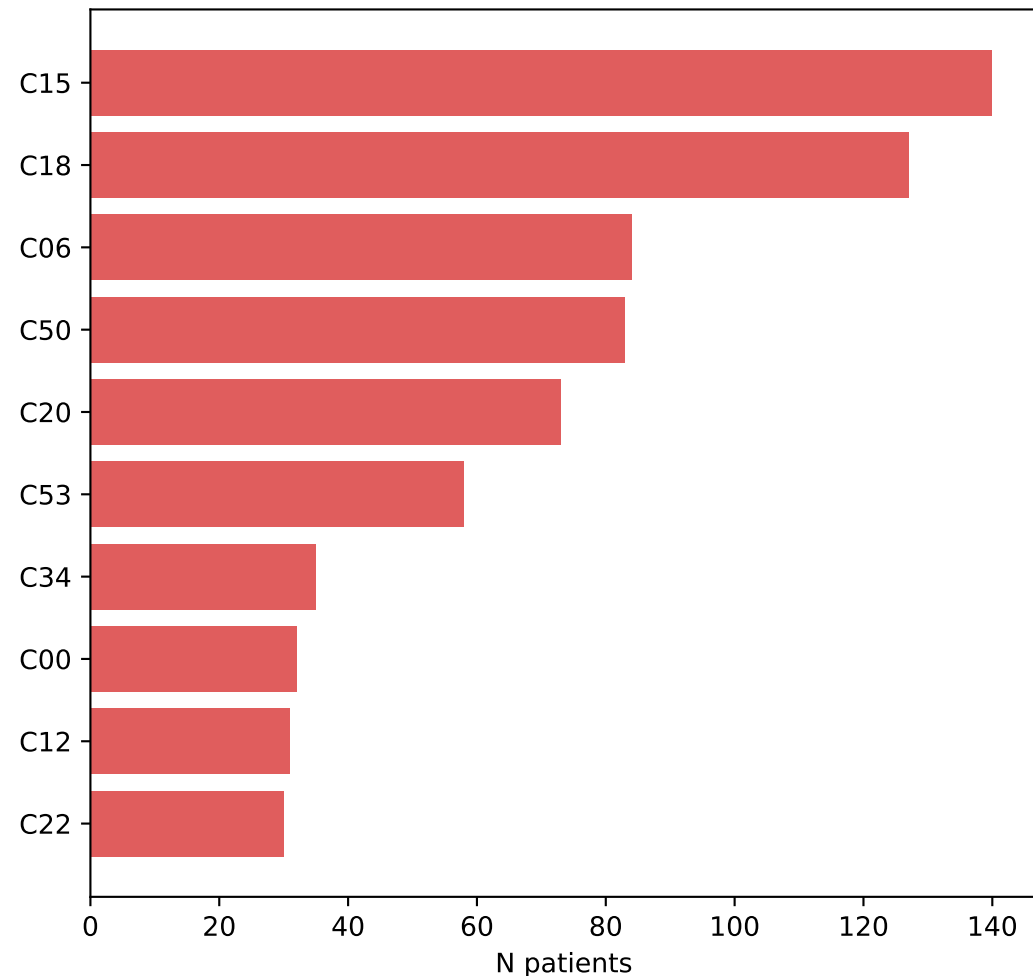
Incremental context value — R@1 vs context length

Incremental value of cancer history for next-cancer prediction
Does each additional cancer in context improve prediction accuracy?

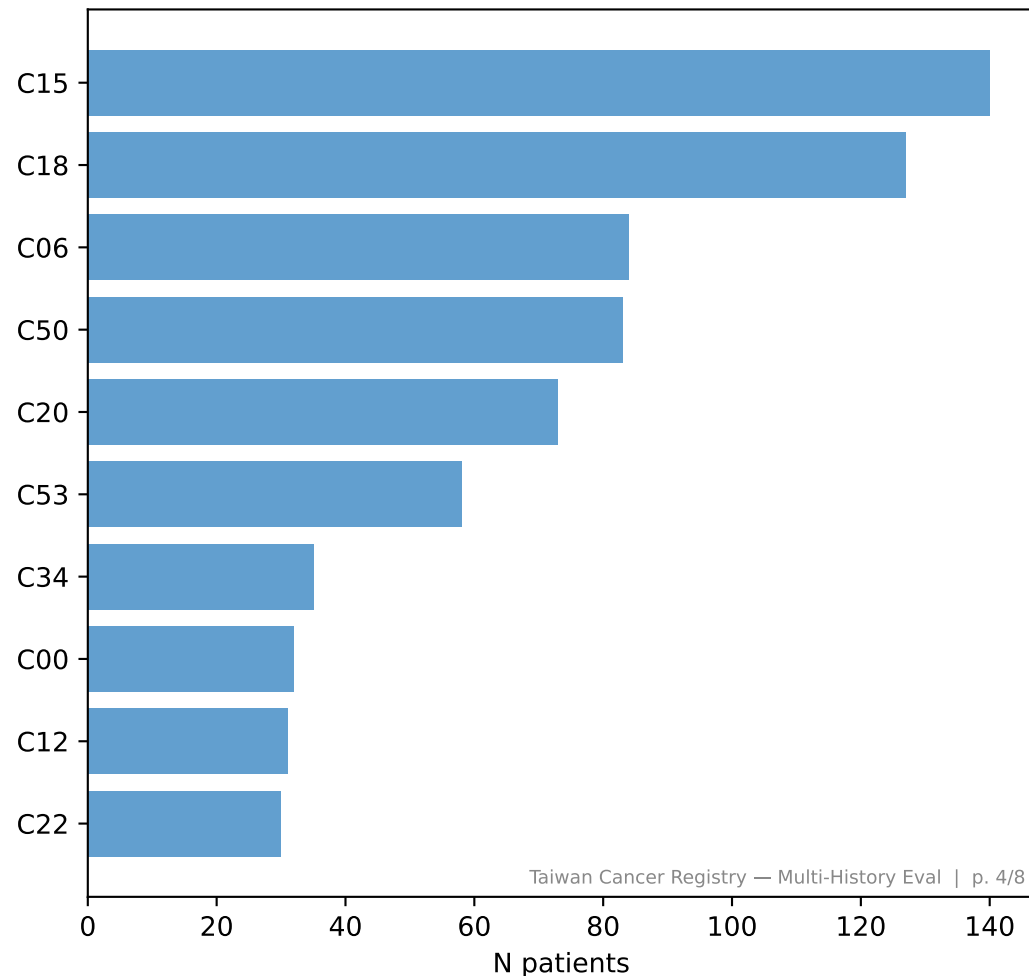


Prediction distribution comparison: Script 25 (causal) vs Script 12 (potentially leaked val split)

Script 25 (k=1 causal): pred1 distribution
(C18 dominant → model collapsed to marginal)



Script 12 (different val split): pred1 distribution
(C15 dominant, site-specific patterns — val split leakage)



Training vs Inference: The Mismatch

TRAINING (Script 07) – Leave-one-out / BERT-style

Sequence: C06 C15 C13 C06
Mask: C06 [MASK] C13 C06 ← random position
Model sees: C06 ??? C13 C06 ← BOTH past AND future
Model learns: bidirectional co-occurrence patterns

R@1 (val, full bidirectional context) = 0.312

CAUSAL INFERENCE (what clinicians need) – left-to-right

At time of 2nd cancer: know C06
Query: C06 [MASK] ____ ← only PAST available
Model sees: C06 ??? <unknown> <unknown>
Must predict WITHOUT future context

R@1 (val, first-only causal context) = 0.067 ← 4.7× worse

FIX: Causal / Autoregressive Masking

Use a causal attention mask (lower-triangular) during training:
Position k can only attend to positions 1..k
This is GPT-style training, not BERT-style

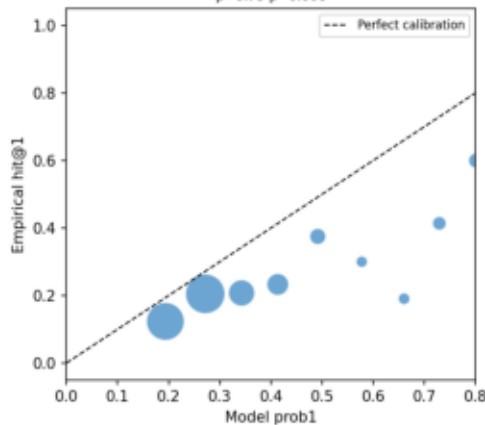
Alternative: Keep BERT training but add causal inference fine-tuning:
Fine-tune with the specific first-cancer-only inference scenario

Model probability calibration — causal inference setting

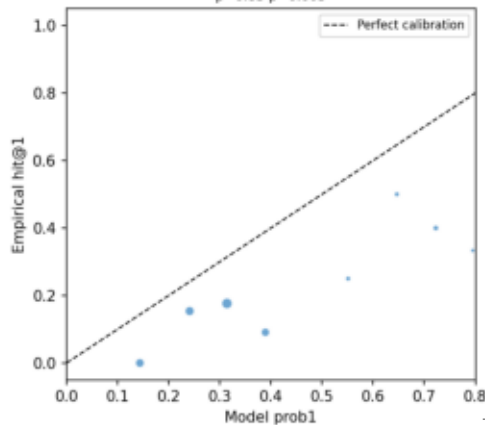
Probability calibration by context length

Probability calibration by context length
(bubble size = n patients in bin)

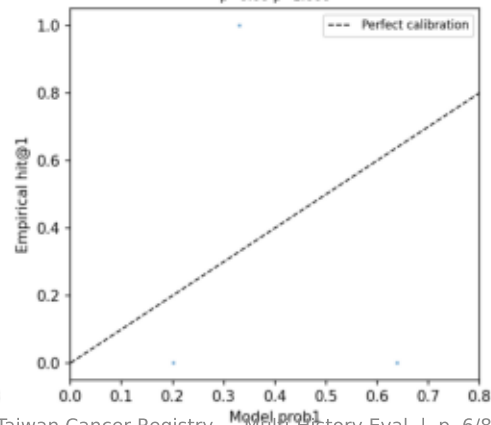
Context k=1 (n=810)
 $\rho=0.79$ $p=0.006$



Context k=2 (n=78)
 $\rho=0.83$ $p=0.005$



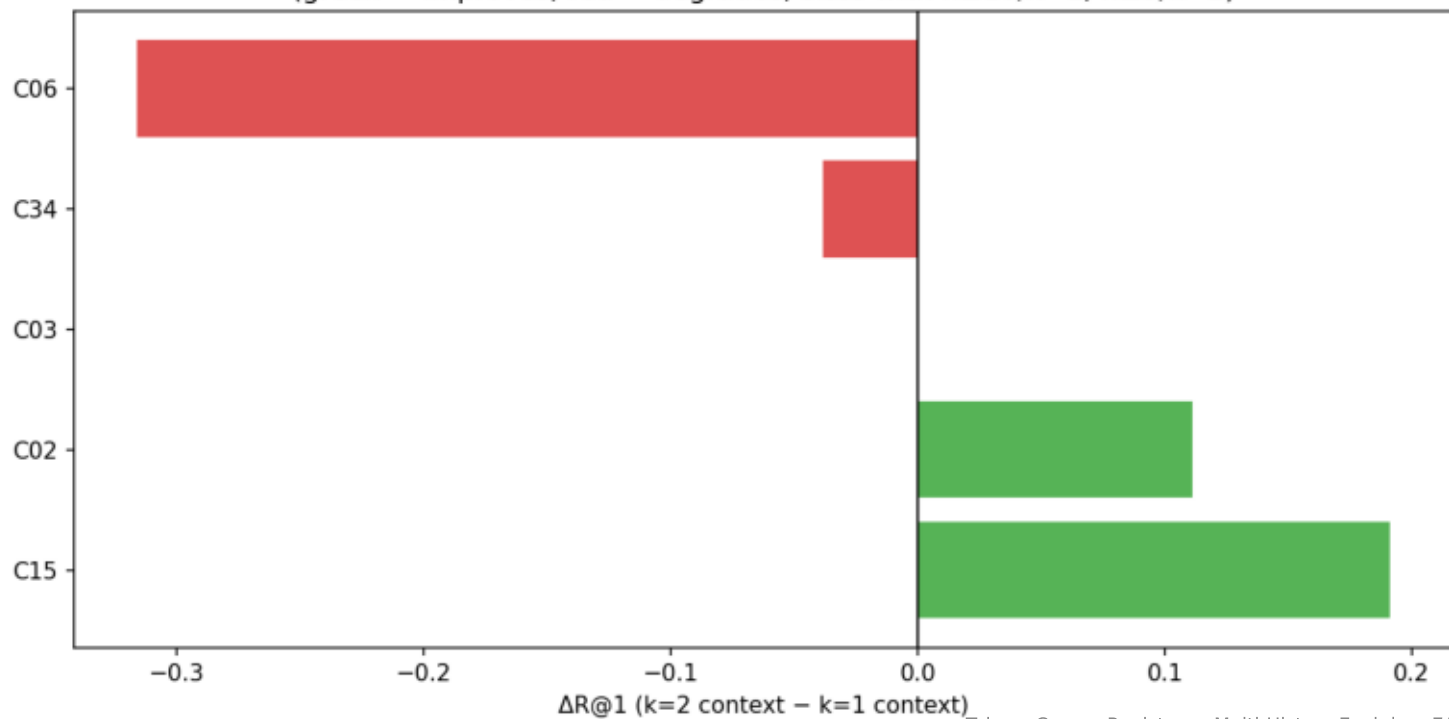
Context k=3 (n=12)
 $\rho=0.00$ $p=1.000$



Per-site accuracy comparison: 2-cancer vs 1-cancer context

Per-site R@1: k=2 vs k=1 context

Per-site R@1 improvement: 2-cancer context vs 1-cancer context
(green = improves; red = degrades; sites with $n \geq 10/k=1$, $n \geq 5/k=2$)



Conclusions and Next Steps

Multi-History Transformer Extension — Taiwan Cancer Registry

Conclusions

1. The Transformer's $R@1=0.312$ (Script 08) is NOT a causal sequential prediction – it is a bidirectional co-occurrence measure where all cancers serve as context simultaneously.
2. In clinically realistic causal inference (predict next from past):
 - $k=1$ context $R@1=0.067$ ($2.5\times$ random, $n=975$)
 - Adding more history does NOT improve performance
 - Model predicts C18 (colon) as top-1 for ~40% of patients regardless of first cancer type → marginal distribution collapse
3. The Script 12 surveillance calendar $R@1=0.232$ reflects a different val split (810 patients from a differently-filtered build_sequences reconstruction). The two val sets have only 18% overlap; the discrepancy is attributable to both different patient selection and potential train-set leakage.
4. This is a fundamental training/inference mismatch. The BERT leave-one-out objective is excellent for learning associations between cancer sites but does not train the model to predict forward from an incomplete sequence.

Recommendations

SHORT TERM – Use Transformer as association model only:

- Report co-occurrence OR / SIR (Scripts 15-16) for clinical guidance
- Use site-specific SIR as the clinical 'risk of second cancer' estimate
- Abandon first-only-context prediction as a clinical tool

MEDIUM TERM – Retrain with causal masking:

- Replace attention mask with lower-triangular (causal) mask
- Train with objective: given first k cancers, predict $k+1$
- Expected improvement: $R@1$ should recover to 0.15-0.25 range
- Evaluate with both first-only and multi-history contexts

LONG TERM – Proper generative model:

- Train an autoregressive sequence model (GPT-style) on cancer sequences sorted by date
- Conditioning on sex, age, birth cohort as prefix tokens
- Sample trajectories for probabilistic clinical planning
- External validation: SEER / UK Biobank multi-primary registry